

PRATHIK ANAND KRISHNAN

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CAREER SUMMARY

Low-latency systems engineer with 4 years of experience building performance-critical C++ infrastructure for trading and market data systems. Experienced in building order execution management systems (OEMS), real-time exchange connectivity and concurrency-sensitive components for algorithmic trading engines. Won IMC's Low-Latency Challenge 2026, breaking the existing IMC record.

Currently pursuing an MSc in Software Engineering at the University of Amsterdam (expected Jul 2026).

WORK EXPERIENCE

SOFTWARE ENGINEER INTERN

03/2026 – 07/2026

TNO-ESI, Eindhoven, Netherlands

- Built a C++ analysis tool for large-scale cyber-physical systems of Philips for my master's thesis. The tool **fuses static and runtime execution traces** to synthesize **SysMLv2 interaction diagrams**, using Graph Driven Intelligence (GDI) techniques with LLMs to reason over event-driven system behaviour.

SENIOR BACKEND ENGINEER, Full-time – Remote

05/2025 – 12/2025

GoQuant – Miami, FL, USA

- Designed and built a **low-latency OEMS in C++17** using Boost.Asio with strand-per-gateway serialization, connecting to 4 exchange gateways (BitMEX, Bitfinex, Deribit, OKX) to support **10+ order types** (TWAP, VWAP, IOC, FOK, etc).
- Migrated market-data serialization from JSON to FlatBuffers, achieving **40-60x decode latency reduction**. Decision derisked by benchmarking under realistic load using **perf stat** before full rollout.
- Diagnosed and resolved a production **p99 latency spike from ~15ms to under 5ms** using perf record and perf annotate and tracing the root cause to hash-map rehashing on the hot path and replaced it with a pre-sized flat structure.
- Led a team of 3 members to develop **4 DEX adapters** (Uniswap v4, Raydium, Orca and Hyperliquid) enabling real-time trade execution, position tracking, and AMM simulations. Used optimized transaction construction to **reduce avg block confirmation time by ~40%** (from 800ms to 480ms).
- Built off-chain **AMM and Blockchain smart contract** interaction layer for Uniswap v4 and Orca DEX, supporting on-chain swaps via off-chain transaction signing.

BACKEND ENGINEER, Full-time – Remote

10/2024 – 04/2025

GoQuant – Miami, FL, USA

- Built real-time **portfolio risk and execution-quality dashboards** using Redis, PostgreSQL and FastAPI for metrics such as **PnL, latency, fill rates and slippage** across all the exchange gateways.
- Implemented **C++ WebSocket server** for multi-client symbol subscriptions **with real-time order book streaming**.
- Developed automated builds and testing scripts using Meson, Conan and Shell scripts achieving **end-to-end CI/CD integration and pipeline automation**.

SOFTWARE DEVELOPER – G3

04/2023 – 10/2024

ARUP – Hyderabad, India

- Designed and shipped a C++20 static library using the **Builder Pattern with CRTP** for type-safe fluent interfaces. Packaged as a NuGet artifact and consumed across 3 proprietary applications.
- Decoupled blocking operations from the critical path using a **mutex-protected worker-thread pipeline** to eliminate latency stalls and ensured data consistency under concurrent access.
- Maintained **96% test coverage** across the SDK using Google Test with strict TDD; integrated SonarQube and CodeCov for static analysis and coverage using **automated build and deployment workflows** via Azure Pipelines.

ARUP – Hyderabad, India

- Created UML and Flow diagrams on **MIRO boards** for low-level and high-level system architecture visualization.
- Participated in **pair programming and Tech-spikes** to address technical debt.
- Built a C++20 event **telemetry system** integrating PostHog via REST API, enabling real-time **feature flag** management and usage analytics across the application.
- I was also hosting **Monthly Software Development Meets (OSM)** at ARUP, creating a platform for knowledge sharing and was contributing to our team's technical growth.

SELECTED ACHIEVEMENTS

- **1st place, IMC Accelerate Low-Latency Challenge 2026 (broke existing IMC record)** | 7th in Tech race | IMC Trading HQ, Amsterdam. Gained firsthand exposure to IMC's low-latency systems, FPGAs, market making strategies and HFT architecture through an intensive two-day engineering program.
- 1st place in **The ADHOC Algorithm Engineering Contest 2026 at TU/e** using multithreaded simulated annealing.
- [MERIT 2022 winner](#) (Global Engineering Competition conducted by ICE – UK)
- Certifications in [Mastering Advanced C++17](#), [Harvard CS50](#), [Meta full-stack development with JavaScript and React](#) and [Duke University – Java Programming](#) in Coursera.

EDUCATION

- Masters in Software Engineering – University of Amsterdam | 2025 - 2026 | 8.2 / 10.0
- Masters in Structural Engineering – BITS Pilani, India | 2018 - 2020 | 9.04 / 10.0

NOTABLE PROJECTS

- [Order Execution System to trade on Deribit using C++17](#): Built a production OMS using Boost.Asio with async WebSocket and REST handlers, supporting order execution, real-time market data streaming, and multi-client order book subscriptions.
- [VanillaVision - Twin Pricing Engine \(C++17\)](#): Built a multi-threaded options pricing engine benchmarking Black-Scholes analytical pricing against Monte Carlo simulation across 10,000 contracts. Achieved 3x throughput improvement over single-threaded baseline by parallelizing simulation paths.
- [Black-Scholes Option Pricing Model Web App \(Python\)](#): This project provides an interactive Black-Scholes Option Pricing Model Web App that visualizes Call and Put option prices on a dynamic heatmap.
- [Other personal projects in C++17 and related web applications](#)

SKILLS & TECHNOLOGIES

Technical Skills & Competencies	Technologies & Tools
- Web & Backend Microservice Development	- C++11,17,20, Python, Rust (learning), TypeScript
- Systems Engineering	- Multithreading, Asynchronous I/O, Event-Driven Architecture, STL, Boost.Asio, Boost.beast, Drogon
- Performance Engineering	- Memory layout, latency optimization, thread safety, profiling
- Trading Systems	- Centralized Exchange OEMS, Market-Data Pipelines, Order Books
- Networking & Protocols	- REST, WebSockets, Protobufs, FlatBuffers, JSON-RPC, FIX
- Build & Integration toolchain, CI/CD	- GCC, MSVC, Meson, Conan, Azure pipelines
- Operating Systems familiarity	- Linux, Windows
- Testing, Automation & Profiling	- GoogleTests, Catch2, PyTest, Python Scripts, TDD, - Vtune, Valgrind, perf stat/record/c2c/annotate
- Databases	- PostgreSQL, Redis, MySQL
- Agile delivery and Scrum processes	- Git, JIRA, MIRO board, Figma